RadixMaths

Practice Paper 2: Mark Scheme and Worked Solutions

Pure Mathematics • Level: Hard

How to use this mark scheme

Each question has a **mark scheme** in the style the exam boards use, then a **worked solution** that explains each step. Mark your own work with the scheme first, then read the worked solution for anything you missed.

M	Method mark: a correct method applied to the problem, even if there are small slips.
dM	Dependent method mark: needs the method mark before it.
A	Accuracy mark: a correct answer from correct working. Needs the method mark it follows.
B	Independent mark for a correct result or statement.
ft	Follow through: allows an earlier wrong answer to be used correctly.
cso	Correct solution only: every step must be correct, with no errors.
awrt	Answers which round to the value given.
oe	Or equivalent: any correct equivalent form is accepted.

For "show that" questions the answer is given, so every step must be shown for full marks. Accept any correct method that is not in the scheme, and give the marks it would earn.

Question 1

Part	Scheme	Marks	Notes
	$2\,(2^x)^2 - 9\,(2^x) + 4 = 0$	M1	Uses $2^{2x+1} = 2\,(2^x)^2$
	$(2\cdot 2^x - 1)(2^x - 4) = 0 \Rightarrow 2^x = \frac{1}{2}$ or 4	M1	Solves the quadratic in 2^x
	$x = -1,\ x = 2$	A1	Both, with working shown
		(3)	

Worked solution

Let $y = 2^x$. Then $2^{2x+1} = 2 \times 2^{2x} = 2\,(2^x)^2 = 2y^2$, so the equation becomes

$$2y^2 - 9y + 4 = 0 \implies (2y-1)(y-4) = 0 \implies y = \tfrac{1}{2} \text{ or } y = 4$$

$2^x = \frac{1}{2} = 2^{-1}$ gives $x = -1$, and $2^x = 4 = 2^2$ gives $x = 2$.

Watch out: 2^{2x+1} is $2 \times (2^x)^2$, not $(2^x)^2 + 1$ or $2^{2x} \times 1$.

Question 2

Part	Scheme	Marks	Notes
(a)	$8^{1/3}\left(1 - \frac{3x}{8}\right)^{1/3} = 2\left(1 - \frac{3x}{8}\right)^{1/3}$	B1	Takes out the factor 2
	$1 + \frac{1}{3}\left(-\frac{3x}{8}\right) + \dfrac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2!}\left(-\frac{3x}{8}\right)^2$	M1	Correct binomial structure for $n = \frac{1}{3}$
	One correct term after multiplying by 2	A1	
	$2 - \dfrac{x}{4} - \dfrac{x^2}{32}$	A1	All correct
(b)	$\lvert x\rvert < \frac{8}{3}$	B1	oe, $-\frac{8}{3} < x < \frac{8}{3}$
(c)	1.997497	B1	cao
		(6)	

Worked solution

(a) The expansion of $(1+u)^n$ needs a 1 at the front, so take out $8^{1/3} = 2$:

$$(8 - 3x)^{\frac{1}{3}} = 2\left(1 - \tfrac{3x}{8}\right)^{\frac{1}{3}}$$

With $n = \frac{1}{3}$ and $u = -\frac{3x}{8}$:

$$(1+u)^{\frac{1}{3}} = 1 + \tfrac{1}{3}u + \frac{\frac{1}{3}\left(-\frac{2}{3}\right)}{2}u^2 + \cdots = 1 + \tfrac{1}{3}u - \tfrac{1}{9}u^2 + \ldots$$

$$= 1 - \frac{x}{8} - \frac{1}{9}\cdot\frac{9x^2}{64} + \cdots = 1 - \frac{x}{8} - \frac{x^2}{64} + \ldots$$

Multiplying by 2 gives $2 - \dfrac{x}{4} - \dfrac{x^2}{32}$.

(b) Valid when $\left\lvert\frac{3x}{8}\right\rvert < 1$, so $\lvert x\rvert < \frac{8}{3}$.

(c) $x = 0.01$ makes $8 - 3x = 7.97$:

$$\sqrt[3]{7.97} \approx 2 - 0.0025 - 0.000003125 = 1.997496875 = 1.997497 \text{ (6 d.p.)}$$

A calculator gives 1.99749686 . . ., so the approximation is correct to 6 decimal places.

Watch out: the factor $8^{1/3} = 2$ multiplies *every* term, not just the first.

Question 3

Part	Scheme	Marks	Notes
	$f'(x) = \lim_{h\to 0} \dfrac{\cos(x+h) - \cos x}{h}$	B1	Correct first-principles set-up
	$\dfrac{\cos x \cos h - \sin x \sin h - \cos x}{h}$	M1	Expands $\cos(x+h)$
	$= \cos x \left(\dfrac{\cos h - 1}{h}\right) - \sin x \left(\dfrac{\sin h}{h}\right)$	M1	Groups into the given limits
	As $h \to 0$, this $\to \cos x \times 0 - \sin x \times 1$	A1	Applies both limits
	So the derivative of $\cos x$ is $-\sin x$	A1	cso, with limit notation used correctly
		(5)	

Worked solution

By definition, the derivative of $f(x) = \cos x$ is

$$f'(x) = \lim_{h\to 0} \frac{\cos(x+h) - \cos x}{h}$$

Expand $\cos(x+h) = \cos x \cos h - \sin x \sin h$ and group the terms so the given limits appear:

$$\frac{\cos x \cos h - \sin x \sin h - \cos x}{h} = \cos x \left(\frac{\cos h - 1}{h}\right) - \sin x \left(\frac{\sin h}{h}\right)$$

As $h \to 0$, $\frac{\cos h - 1}{h} \to 0$ and $\frac{\sin h}{h} \to 1$, so

$$f'(x) = \cos x \times 0 - \sin x \times 1 = -\sin x$$

Watch out: write "$\lim_{h\to 0}$" or "as $h \to 0$" at the right moment. Don't put $h = 0$ into the fraction, because you would be dividing by zero.

Question 4

Part	Scheme	Marks	Notes
(a)	$\dfrac{a}{1-r} = 12$ and $\dfrac{a(1-r^3)}{1-r} = \dfrac{21}{2}$	M1	Both formulae used
	$12(1-r^3) = \frac{21}{2} \Rightarrow r^3 = \frac{1}{8}$	M1	Eliminates a
	$r = \frac{1}{2}$, $a = 6$	A1	Both
(b)	$12\left(\frac{1}{2}\right)^n < 0.001 \Rightarrow n > \log_2 12000 = 13.55\ldots$	M1	Uses $S_\infty - S_n = \frac{ar^n}{1-r}$ and logs
	$n = 14$	A1	
		(5)	

Worked solution

(a) $S_\infty = \dfrac{a}{1-r} = 12$, and $S_3 = \dfrac{a(1-r^3)}{1-r} = S_\infty(1-r^3)$. So

$$12(1-r^3) = \frac{21}{2} \Rightarrow 1-r^3 = \frac{7}{8} \Rightarrow r^3 = \frac{1}{8} \Rightarrow r = \frac{1}{2}$$

Then $a = 12(1-\frac{1}{2}) = 6$.
(b) The difference between S_∞ and S_n is

$$S_\infty - S_n = \frac{a}{1-r} - \frac{a(1-r^n)}{1-r} = \frac{ar^n}{1-r} = 12\left(\tfrac{1}{2}\right)^n$$

Solve $12\left(\frac{1}{2}\right)^n < 0.001$: $2^n > 12\,000$, so $n > \dfrac{\ln 12\,000}{\ln 2} = 13.55\ldots$. The smallest integer is $n = 14$.
Check: $n = 13$ gives 0.00146, which is too big, and $n = 14$ gives 0.00073.

Question 5

Part	Scheme	Marks	Notes
(a)	$\dfrac{2(x^2+2)-(2x-1)(2x)}{(x^2+2)^2}$	M1	Quotient rule, correct order
	$= \dfrac{-2x^2+2x+4}{(x^2+2)^2}$	A1	
	$= \dfrac{-2(x-2)(x+1)}{(x^2+2)^2}$	A1*	Given answer
(b)	$f'(x) = 0 \Rightarrow x = 2,\ x = -1$	M1	
	$\left(2, \frac{1}{2}\right)$	A1	
	$(-1, -1)$	A1	
(c)	Only two turning points, and $f(x) \to 0$ as $x \to \pm\infty$	M1	Some justification
	$-1 \leqslant f(x) \leqslant \frac{1}{2}$	A1	Inclusive inequalities
		(8)	

Worked solution

(a) Quotient rule with $u = 2x-1$, $v = x^2+2$, so $u' = 2$ and $v' = 2x$:

$$f'(x) = \frac{2(x^2+2)-(2x-1)(2x)}{(x^2+2)^2} = \frac{2x^2+4-4x^2+2x}{(x^2+2)^2} = \frac{-2(x^2-x-2)}{(x^2+2)^2} = \frac{-2(x-2)(x+1)}{(x^2+2)^2}$$

(b) The denominator is never 0, so $f'(x) = 0$ when $x = 2$ or $x = -1$. $f(2) = \frac{3}{6} = \frac{1}{2}$ and $f(-1) = \frac{-3}{3} = -1$. The points are $\left(2, \frac{1}{2}\right)$ and $(-1, -1)$.
(c) As $x \to \pm\infty$, $f(x) \to 0$, because the x^2 in the denominator grows faster than the numerator. The curve is continuous and these are its only turning points, so it never goes above the maximum $\frac{1}{2}$ or below the minimum -1:

$$-1 \leqslant f(x) \leqslant \tfrac{1}{2}$$

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Question 6

Part	Scheme	Marks	Notes
	$\overrightarrow{OM} = \frac{1}{2}(\mathbf{a}+\mathbf{b})$	B1	
	$\overrightarrow{OQ} = \mathbf{a} + \lambda\left(\frac{2}{3}\mathbf{b} - \mathbf{a}\right)$	M1	Route via A along AP
	$\overrightarrow{OQ} = \mu \cdot \frac{1}{2}(\mathbf{a}+\mathbf{b})$; compare coefficients of $\mathbf{a}$ and $\mathbf{b}$	M1	Two equations
	$\lambda = \frac{3}{5},\ \mu = \frac{4}{5}$	A1	
	$OQ : QM = 4 : 1$	A1	
		(5)	

Worked solution

$\overrightarrow{OP} = \frac{2}{3}\mathbf{b}$, so $\overrightarrow{AP} = \frac{2}{3}\mathbf{b} - \mathbf{a}$, and $\overrightarrow{OM} = \frac{1}{2}(\mathbf{a}+\mathbf{b})$.
Q lies on AP, so $\overrightarrow{OQ} = \mathbf{a} + \lambda\left(\frac{2}{3}\mathbf{b} - \mathbf{a}\right) = (1-\lambda)\mathbf{a} + \frac{2\lambda}{3}\mathbf{b}$ for some λ.
Q also lies on OM, so $\overrightarrow{OQ} = \frac{\mu}{2}\mathbf{a} + \frac{\mu}{2}\mathbf{b}$ for some μ.
$\mathbf{a}$ and $\mathbf{b}$ are not parallel, so the coefficients must match:

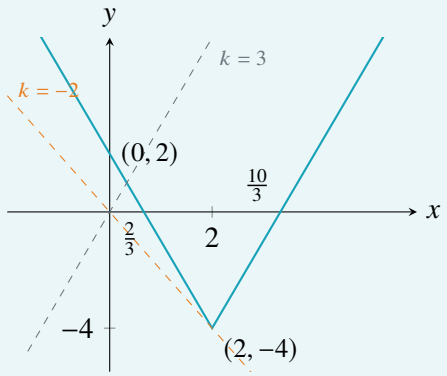
$$1 - \lambda = \frac{\mu}{2}, \qquad \frac{2\lambda}{3} = \frac{\mu}{2} \Rightarrow 1 - \lambda = \frac{2\lambda}{3} \Rightarrow \lambda = \frac{3}{5},\ \mu = \frac{4}{5}$$

So $\overrightarrow{OQ} = \frac{4}{5}\overrightarrow{OM}$, and $OQ : QM = 4 : 1$.

Question 7

Part	Scheme	Marks	Notes
(a)	V shape with vertex at $(2, -4)$	B1	
	y-intercept $(0, 2)$	B1	
	x-intercepts $\left(\frac{2}{3}, 0\right)$ and $\left(\frac{10}{3}, 0\right)$	B1	
(b)	$3(x-2) - 4 = \frac{1}{2}x + 1$ or $3(2-x) - 4 = \frac{1}{2}x + 1$	M1	One correct branch
	$x = \frac{22}{5}$	A1	
	$x = \frac{2}{7}$	A1	No extra solutions
(c)	$y = kx$ through the vertex: $k = -2$	M1	
	Parallel to the right-hand branch: $k = 3$	M1	
	Both critical values -2 and 3	A1	
	$\{k : -2 < k < 3\}$	A1	Strict inequalities
		(10)	

Worked solution



(a) $y = 3|x-2| - 4$ is $y = |x|$ moved right 2, stretched by 3 vertically, then moved down 4. The vertex is $(2, -4)$. At $x = 0$, $y = 6 - 4 = 2$. $y = 0$ when $|x-2| = \frac{4}{3}$, so $x = \frac{2}{3}$ or $x = \frac{10}{3}$.
(b) Solve on each branch, and keep a solution only if it lies on that branch.

- Right branch ($x \geqslant 2$): $3x - 10 = \frac{1}{2}x + 1 \Rightarrow \frac{5}{2}x = 11 \Rightarrow x = \frac{22}{5}$. Valid, since $\frac{22}{5} \geqslant 2$.
- Left branch ($x < 2$): $2 - 3x = \frac{1}{2}x + 1 \Rightarrow 1 = \frac{7}{2}x \Rightarrow x = \frac{2}{7}$. Valid, since $\frac{2}{7} < 2$.

(c) The solutions are where the graph meets the line $y = kx$, which always passes through the origin. Rotate this line about O (see the dashed lines above):

- At $k = -2$ the line passes through the vertex $(2, -4)$: one solution.
- For k just above -2, the line crosses the left branch and the right branch: two solutions.
- As k increases towards 3, the line still meets both branches. At $k = 3$ it is parallel to the right branch $y = 3x - 10$, so only the left crossing remains: one solution. For $k > 3$ there is also only one.
- For $-3 \leqslant k < -2$ the line misses the graph, and for $k < -3$ it crosses only the left branch.

Exactly two solutions: $\{k : -2 < k < 3\}$.

Question 8

Part	Scheme	Marks	Notes
(a)	$R = 13$	B1	
	$\tan\alpha = \frac{12}{5}$	M1	
	$\alpha = 1.176$	A1	
(b)	$\sin(\theta - 1.176) = \frac{6.5}{13} = \frac{1}{2}$	M1	
	$\theta - \alpha = \frac{\pi}{6}$ or $\frac{5\pi}{6}$	M1	Finds a second value
	$\theta = 1.70,\ 3.79$	A1	Both, no extras
(c)	Maximum value $\frac{1}{7}$	B1	
	$\sin(\theta - \alpha) = -1 \Rightarrow \theta - \alpha = \frac{3\pi}{2}$	M1	
	$\theta = 5.89$	A1	awrt 5.89
		(9)	

Worked solution

(a) $R\sin(\theta - \alpha) = R\sin\theta\cos\alpha - R\cos\theta\sin\alpha$. Comparing with $5\sin\theta - 12\cos\theta$ gives $R\cos\alpha = 5$ and $R\sin\alpha = 12$. So

$$R = \sqrt{5^2 + 12^2} = 13, \qquad \tan\alpha = \frac{12}{5} \Rightarrow \alpha = 1.176 \text{ (3 d.p.)}$$

(b) $13\sin(\theta-\alpha)=6.5$, so $\sin(\theta-\alpha)=\frac{1}{2}$. Since $-1.176 \leqslant \theta-\alpha < 5.107$, the values are $\theta-\alpha=\frac{\pi}{6}$ and $\frac{5\pi}{6}$. So $\theta=1.70$ and $\theta=3.79$ (2 d.p.).

(c) $\mathrm{g}(\theta)=\dfrac{1}{20+13\sin(\theta-\alpha)}$. This is largest when the denominator is smallest, which happens when $\sin(\theta-\alpha)=-1$. The denominator is then $20-13=7$, so the maximum is $\frac{1}{7}$.

$\theta-\alpha=-\frac{\pi}{2}$ gives a negative θ, so use $\theta-\alpha=\frac{3\pi}{2}$: $\theta=1.176+4.712=5.89$.

Watch out: the denominator is always at least 7, so it is never zero and g is always defined.

Question 9

Part	Scheme	Marks	Notes
	$\frac{du}{dx}=e^x$, so $dx=\dfrac{du}{u-1}$	B1	oe
	$\displaystyle\int \frac{(u-1)^2}{u}\cdot\frac{1}{u-1}\,du$	M1	Fully in terms of u
	$\displaystyle=\int\left(1-\frac{1}{u}\right)du=u-\ln u$	A1	
	Limits $u=2$ and $u=4$	B1	
	$(4-\ln 4)-(2-\ln 2)=2-\ln 2$	A1*	cso, uses $\ln 4=2\ln 2$
		(5)	

Worked solution

$u=e^x+1$, so $e^x=u-1$, $e^{2x}=(u-1)^2$ and $\frac{du}{dx}=e^x=u-1$. The limits change too: $x=0\Rightarrow u=2$, and $x=\ln 3\Rightarrow u=3+1=4$.

$$\int_0^{\ln 3}\frac{e^{2x}}{e^x+1}\,dx=\int_2^4\frac{(u-1)^2}{u}\cdot\frac{du}{u-1}=\int_2^4\frac{u-1}{u}\,du=\int_2^4\left(1-\frac{1}{u}\right)du$$

$$=\Big[u-\ln u\Big]_2^4=(4-\ln 4)-(2-\ln 2)=2-2\ln 2+\ln 2=2-\ln 2$$

Question 10

Part	Scheme	Marks	Notes
(a)	$3x^2+3y^2\dfrac{dy}{dx}$	M1	Implicit differentiation of y^3
	$=6y+6x\dfrac{dy}{dx}$	M1	Product rule on $6xy$
	$\dfrac{dy}{dx}=\dfrac{2y-x^2}{y^2-2x}$	A1	oe
(b)	$2y-x^2=0\Rightarrow y=\frac{x^2}{2}$, substituted into C	M1	
	$x^3+\frac{x^6}{8}=3x^3\Rightarrow x^3=16$	M1	Rejects $x=0$
	$\left(2\sqrt[3]{2},\ 2\sqrt[3]{4}\right)$	A1	oe, such as $\left(\sqrt[3]{16},\sqrt[3]{32}\right)$
		(6)	

Worked solution

(a) Differentiate every term with respect to x. Use the chain rule on y^3 and the product rule on $6xy$:

$$3x^2 + 3y^2\frac{\mathrm{d}y}{\mathrm{d}x} = 6y + 6x\frac{\mathrm{d}y}{\mathrm{d}x} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x}\left(3y^2 - 6x\right) = 6y - 3x^2 \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2y - x^2}{y^2 - 2x}$$

(b) The tangent is parallel to the x-axis when the numerator is zero: $y = \frac{x^2}{2}$. Substitute into $x^3 + y^3 = 6xy$:

$$x^3 + \frac{x^6}{8} = 3x^3 \Rightarrow \frac{x^6}{8} = 2x^3 \Rightarrow x^3(x^3 - 16) = 0$$

$x = 0$ is the origin, so $x^3 = 16$ and $x = \sqrt[3]{16} = 2\sqrt[3]{2}$. Then $y = \frac{x^2}{2} = \frac{4\sqrt[3]{4}}{2} = 2\sqrt[3]{4}$.
Check the denominator there: $y^2 - 2x = 4\sqrt[3]{16} - 4\sqrt[3]{2} = 4\sqrt[3]{2} \neq 0$, so the gradient really is 0.

Question 11

Part	Scheme	Marks	Notes
(a)	$\dfrac{1-(1-2\sin^2\theta)}{\sin 2\theta}$	M1	Uses $\cos 2\theta = 1 - 2\sin^2\theta$
	$= \dfrac{2\sin^2\theta}{2\sin\theta\cos\theta}$	M1	Uses $\sin 2\theta = 2\sin\theta\cos\theta$
	$= \tan\theta$	A1*	cso
(b)	$\tan 2x = 3\tan x$	M1	Uses (a) with $\theta = 2x$
	$\dfrac{2\tan x}{1-\tan^2 x} = 3\tan x \Rightarrow \tan x\,(3\tan^2 x - 1) = 0$	M1	
	$\tan x = \pm\frac{1}{\sqrt{3}}$ ($\tan x = 0$ gives no solution in range)	A1	
	$x = \frac{\pi}{6}, \frac{5\pi}{6}$	A1	Both, no extras
		(7)	

Worked solution

(a)

$$\frac{1-\cos 2\theta}{\sin 2\theta} = \frac{1-(1-2\sin^2\theta)}{2\sin\theta\cos\theta} = \frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \frac{\sin\theta}{\cos\theta} = \tan\theta$$

(b) Using (a) with $\theta = 2x$, the left-hand side is $\tan 2x$. So

$$\frac{2\tan x}{1-\tan^2 x} = 3\tan x \Rightarrow 2\tan x = 3\tan x - 3\tan^3 x \Rightarrow \tan x\left(3\tan^2 x - 1\right) = 0$$

- $\tan x = 0$ gives $x = 0$ or π, but neither is in $0 < x < \pi$.
- $\tan x = \frac{1}{\sqrt{3}}$ gives $x = \frac{\pi}{6}$, and $\tan x = -\frac{1}{\sqrt{3}}$ gives $x = \frac{5\pi}{6}$.

Check that $\sin 4x \neq 0$ for both: $\sin\frac{2\pi}{3} = \frac{\sqrt{3}}{2}$ and $\sin\frac{10\pi}{3} = -\frac{\sqrt{3}}{2}$, so both are valid. The solutions are $x = \frac{\pi}{6}, \frac{5\pi}{6}$.

Watch out: don't divide by $\tan x$, or you lose the factor that you then need to check.

Question 12

Part	Scheme	Marks	Notes
(a)	$h = 0.5$; y values $0,\ 0.4055,\ 0.6931,\ 0.9163,\ 1.0986$	B1	
	$\frac{0.5}{2}\,[0 + 1.0986 + 2(0.4055 + 0.6931 + 0.9163)]$	M1	Correct structure
	1.282	A1	
(b)	$u = \ln x,\ \frac{\mathrm{d}v}{\mathrm{d}x} = 1$: $x\ln x - \int 1\,\mathrm{d}x$	M1	
	$[x\ln x - x]_1^3$	A1	
	$3\ln 3 - 2$	A1	
(c)	$\ln x$ is concave, so each chord lies below the curve	B1	oe, $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{1}{x^2} < 0$
		(7)	

Worked solution

(a) Four strips from 1 to 3 gives $h = 0.5$ and $x = 1, 1.5, 2, 2.5, 3$:

$$\int_1^3 \ln x\,\mathrm{d}x \approx \frac{0.5}{2}\,[\ln 1 + \ln 3 + 2(\ln 1.5 + \ln 2 + \ln 2.5)] = 0.25 \times 5.1284\ldots = 1.282$$

(b) Write $\ln x$ as $1 \times \ln x$ and use parts with $u = \ln x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$, so $v = x$:

$$\int \ln x\,\mathrm{d}x = x\ln x - \int x \cdot \frac{1}{x}\,\mathrm{d}x = x\ln x - x$$

$$\int_1^3 \ln x\,\mathrm{d}x = (3\ln 3 - 3) - (0 - 1) = 3\ln 3 - 2 = 1.2958\ldots$$

(c) $\frac{\mathrm{d}^2}{\mathrm{d}x^2}(\ln x) = -\frac{1}{x^2} < 0$, so the curve is concave. The top of each trapezium is a chord, which lies below a concave curve, so the trapezium rule underestimates. The error here is about 1.1%.

Question 13

Part	Scheme	Marks	Notes
	Assume there are positive integers x, y with $x^2 - y^2 = 10$	B1	Clear assumption
	$(x - y)(x + y) = 10$	M1	
	$x - y$ and $x + y$ have the same parity (they differ by $2y$)	M1	Or factor pairs, or cases on parity
	10 can't be written as a product of two factors with the same parity: 1×10, 2×5	A1	All cases
	Contradiction, so there are no such integers	A1	Needs a conclusion
		(5)	

Worked solution

Assume, for contradiction, that there are positive integers x and y with $x^2 - y^2 = 10$. Then

$$(x - y)(x + y) = 10$$

$x + y$ is a positive integer, so $x - y$ must also be a positive integer, and $x - y < x + y$. The only factor pairs are 1×10 and 2×5.

But $(x+y)-(x-y)=2y$ is even, so $x-y$ and $x+y$ are both odd or both even. In 1×10 and in 2×5, one factor is odd and the other is even. This is a contradiction.
So there are no positive integers x and y with $x^2-y^2=10$.

Another way: if x and y have the same parity, x^2-y^2 is a multiple of 4, but 10 isn't. If they have different parity, x^2-y^2 is odd, but 10 is even.

Question 14

Part	Scheme	Marks	Notes
(a)	$\displaystyle\int \frac{2500}{P(500-P)}\,\mathrm{d}P=\int 1\,\mathrm{d}t$	M1	Separates the variables
	$\displaystyle\frac{2500}{P(500-P)}\equiv\frac{A}{P}+\frac{B}{500-P}$	M1	Partial fractions
	$A=5,\ B=5$	A1	
	$5\ln P-5\ln(500-P)=t+c$	M1	Integrates to two logs
	$t=0,\ P=50\Rightarrow c=5\ln\frac{1}{9}$	M1	Finds the constant
	$\displaystyle\frac{P}{500-P}=\frac{1}{9}\mathrm{e}^{0.2t}\Rightarrow P=\frac{500}{1+9\mathrm{e}^{-0.2t}}$	A1*	cso
(b)	$\displaystyle 250=\frac{500}{1+9\mathrm{e}^{-0.2t}}\Rightarrow \mathrm{e}^{-0.2t}=\frac{1}{9}$	M1	
	$t=5\ln 9=10.99$ years	A1	
(c)	The number of fish approaches 500	B1	
		(9)	

Worked solution

(a) Separate the variables:

$$\int \frac{2500}{P(500-P)}\,\mathrm{d}P=\int 1\,\mathrm{d}t$$

Partial fractions: $2500=A(500-P)+BP$. $P=0$ gives $A=5$, and $P=500$ gives $B=5$. So

$$\int\left(\frac{5}{P}+\frac{5}{500-P}\right)\mathrm{d}P=5\ln P-5\ln(500-P)=t+c$$

Both P and $500-P$ are positive here, so no modulus signs are needed. At $t=0$, $P=50$: $c=5\ln\frac{50}{450}=5\ln\frac{1}{9}$. So

$$5\ln\frac{P}{500-P}=t+5\ln\frac{1}{9}\ \Rightarrow\ \frac{P}{500-P}=\frac{1}{9}\mathrm{e}^{0.2t}$$

$$9P=(500-P)\mathrm{e}^{0.2t}\ \Rightarrow\ P\left(9+\mathrm{e}^{0.2t}\right)=500\mathrm{e}^{0.2t}\ \Rightarrow\ P=\frac{500}{1+9\mathrm{e}^{-0.2t}}$$

Dividing the top and bottom by $\mathrm{e}^{0.2t}$ gives the last step.
(b) $1+9\mathrm{e}^{-0.2t}=2$, so $\mathrm{e}^{-0.2t}=\frac{1}{9}$ and $t=5\ln 9=10.99$ years.
(c) As $t\to\infty$, $\mathrm{e}^{-0.2t}\to0$, so $P\to500$. The number of fish levels off at 500, the most the lake can support according to the model.

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Question 15

Part	Scheme	Marks	Notes
(a)	$\dfrac{dy}{dx} = \dfrac{3t^2-3}{2t}$	M1	$\frac{dy}{dt} \div \frac{dx}{dt}$
	$3t^2 - 3 = 0 \Rightarrow t = \pm 1$	M1	
	$(1, -2)$ and $(1, 2)$	A1	
(b)	$y^2 = t^2(t^2-3)^2$	M1	Squares $y = t(t^2-3)$
	$= x(x-3)^2$	A1*	
(c)	Loop runs from $x = 0$ to $x = 3$	B1	A is $(3, 0)$
	Area $= 2\displaystyle\int_0^3 \sqrt{x}(3-x)\,dx$	M1	Uses symmetry, or the top and bottom halves
	$= 2\left[2x^{3/2} - \frac{2}{5}x^{5/2}\right]$	M1	Integrates the powers
	$= 2\left(6\sqrt{3} - \frac{18\sqrt{3}}{5}\right)$	dM1	Applies limits 0 and 3
	$\dfrac{24\sqrt{3}}{5}$	A1	
		(10)	

Worked solution

(a) $\frac{dx}{dt} = 2t$ and $\frac{dy}{dt} = 3t^2 - 3$, so $\dfrac{dy}{dx} = \dfrac{3t^2-3}{2t}$. This is zero when $t^2 = 1$:

- $t = 1$: $(1, 1 - 3) = (1, -2)$
- $t = -1$: $(1, -1 + 3) = (1, 2)$

(b) $y = t(t^2-3)$, so $y^2 = t^2(t^2-3)^2$. Replace t^2 with x: $y^2 = x(x-3)^2$.

(c) The curve crosses the x-axis when $y = 0$, so $t = 0$ or $t = \pm\sqrt{3}$. These give O and $A(3, 0)$, so the loop lies between $x = 0$ and $x = 3$.

The curve is symmetric in the x-axis, because y^2 is unchanged if y becomes $-y$. For $0 \leqslant x \leqslant 3$, the top half is $y = \sqrt{x}(3-x)$. So

$$\text{Area} = 2\int_0^3 \left(3x^{1/2} - x^{3/2}\right) dx = 2\left[2x^{3/2} - \tfrac{2}{5}x^{5/2}\right]_0^3$$

$3^{3/2} = 3\sqrt{3}$ and $3^{5/2} = 9\sqrt{3}$, so

$$\text{Area} = 2\left(6\sqrt{3} - \frac{18\sqrt{3}}{5}\right) = 2 \times \frac{12\sqrt{3}}{5} = \frac{24\sqrt{3}}{5}$$

Another way, if your course covers areas with parametric equations: $2\displaystyle\int_0^{\sqrt{3}} (3t - t^3)(2t)\,dt = \frac{24\sqrt{3}}{5}$.

End of mark scheme. Every answer here was checked with computer algebra.